CHE 312

1. **<u>Course number and name</u>**
   CHE 312. Transport phenomena II

2. **<u>Credits, contact hours and categorization</u>**
   Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. **<u>Instructor's or course coordinator's name</u>**
   Korosh Torabi, Christos Takoudis

4. **<u>Textbook, title, author, publisher, year</u>**
   Required (KT) –Transport Phenomena, Revised 2nd Edition by R. B. Bird, W. E. Stewart and E. N. Lightfoot. John Wiley & Sons, Inc (2006).
   Required (CT) -- Theodore L. Bergman, Adrienne S. Lavine, *Fundamentals of HEAT and MASS Transfer*, Wiley, Eighth Edition, 2019

5. **<u>Specific course information</u>**
   a. <u>Brief description of the content of the course (catalog description)</u>
   Heat and mass transport phenomena. Heat conduction, convection and radiation. Heat exchanger design. Diffusion. Mass transfer coefficients.
   b. <u>Prerequisites or co-requisites</u>
   CHE 311
   c. <u>Indicate whether a required, elective, or selected elective.</u>
   Required.

6. **<u>Specific goals for the course</u>**
   a. <u>Specific outcomes of instruction</u>.
   The broad objectives of this course are:
   (a) To develop clear understanding of the basic mechanisms of heat and mass transfer in physical processes encountered in daily living experiences and in chemical engineering practice; and
   (b) To develop the skills to describe heat and mass processes in terms of mathematical models, make appropriate assumptions to simplify the models, and perform calculations to obtain qualitative and/or quantitative information which can lead to better design or better understanding of the processes.

a. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3)	an ability to communicate effectively with a range of audiences.	X			
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	X			
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Introduction to Conduction
- One-Dimensional Steady-State Conduction
- Two-Dimensional Steady-State Conduction
- Transient Conduction
- Introduction to Convection
- Heat Transfer in External Flows
- Heat Transfer in Internal Flows
- Free Convection
- Heat Exchangers
- Radiation Heat Transfer
- Radiation Exchange Between Surfaces
- Diffusion Mass Transfer
- Diffusion – Homogeneous Chemical Reaction Systems
- Diffusion – Heterogeneous Chemical Reaction Systems
- Transient Diffusion and Applications
- Diffusion and Chemical Reaction Inside a Porous Catalyst
- Mass Transfer Design Calculations
- Multimode Heat and Mass Transfer